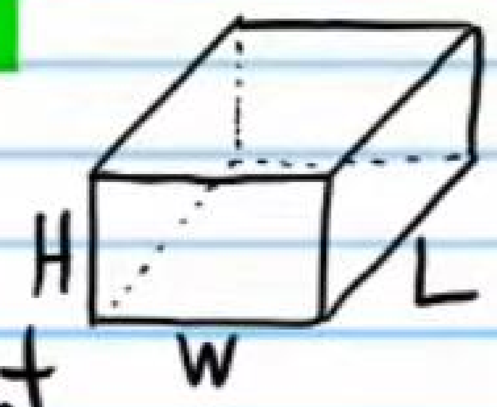
Name
Date
Course

Volume

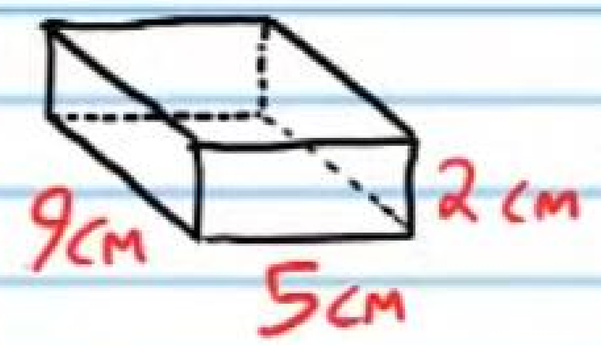
Volume of a Rectangular Prism

$V = LWH$



L = length, W = width, and H = height

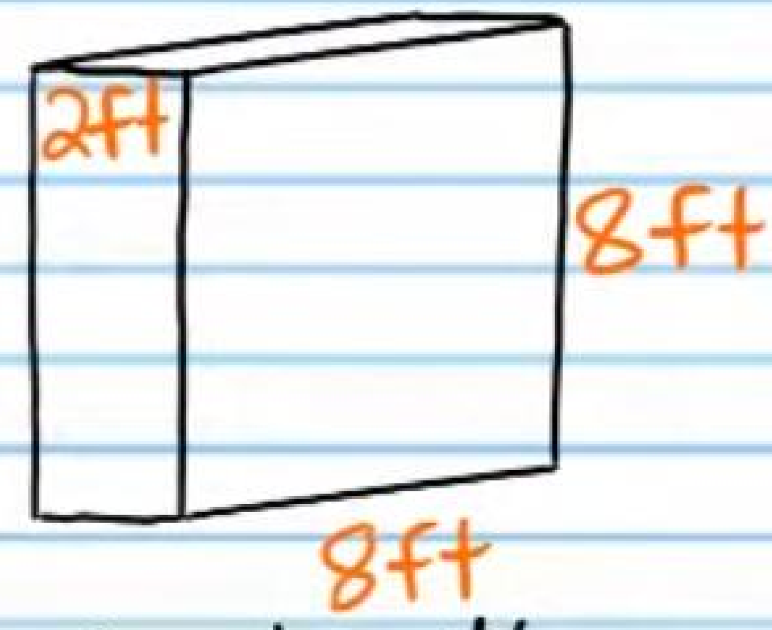
Find the volume of each rectangular prism below.

① 

$V = LWH$

$V = (9)(5)(2)$

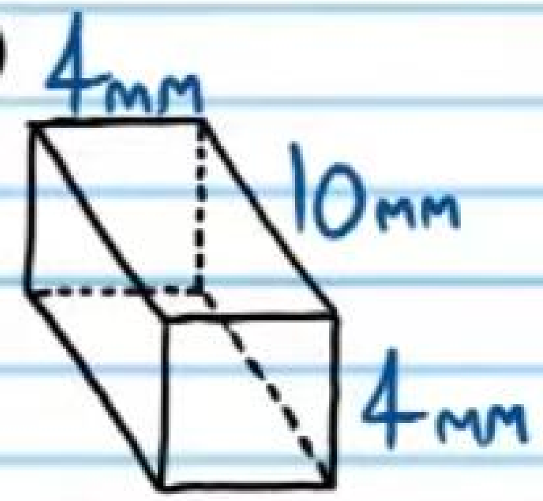
$\boxed{V = 90\ cm^3}$

② 

$V = LWH$

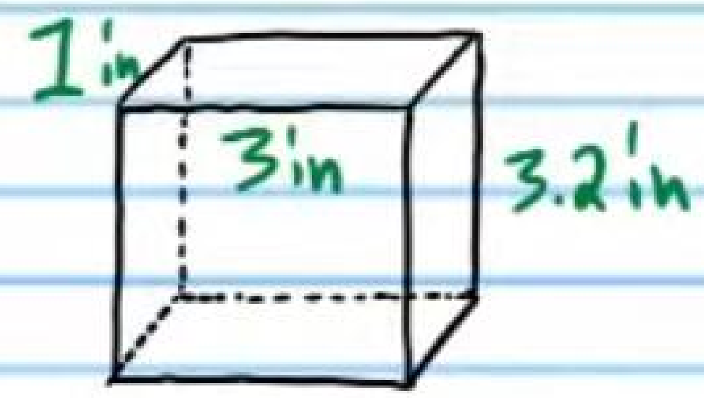
$V = (8)(8)(2)$

$\boxed{V = 128\ ft^3}$

③ 

$V = 4(10)(4)$

$\boxed{V = 160\ mm^3}$

④ 

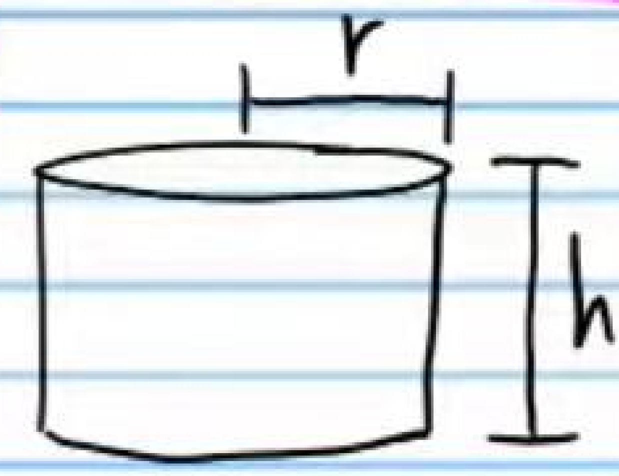
$V = 1(3)(3.2)$

$\boxed{V = 9.6\ in^3}$

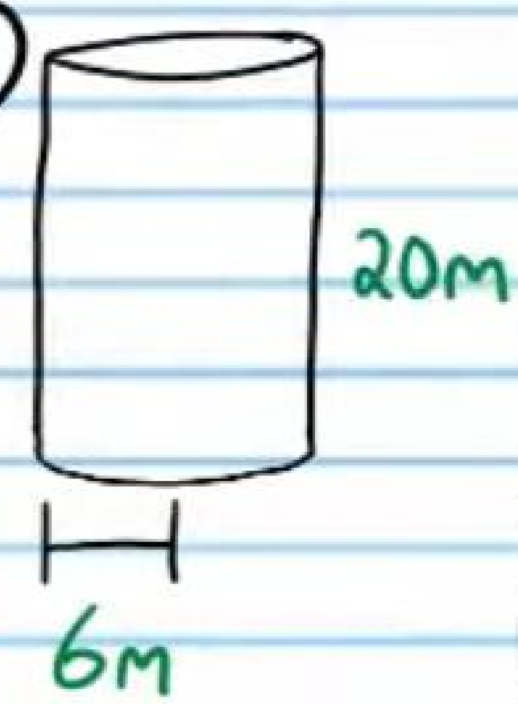
Volume of a Cylinder

$V = \pi r^2 h$

r = radius & h = height



Find the volume of each cylinder below.
Leave each answer as a multiple of π.

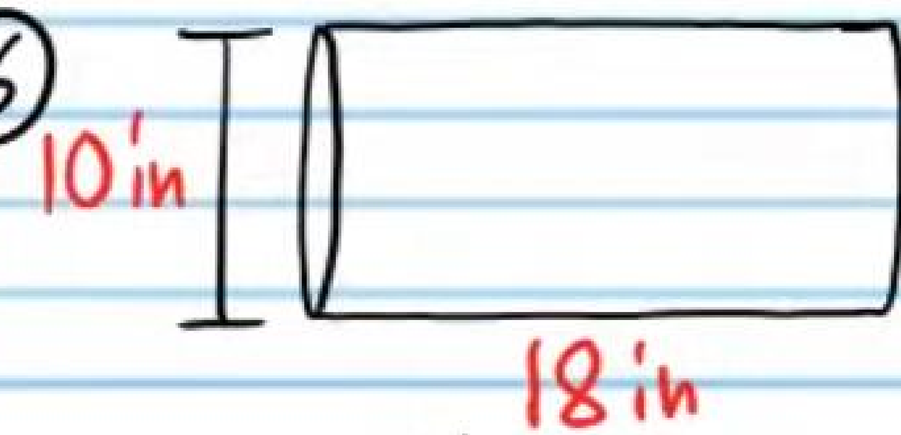
⑤ 

$V = \pi r^2 h$

$V = \pi(6)^2(20)$

$V = \pi(36)20$

$\boxed{V = 720\pi\ m^3}$

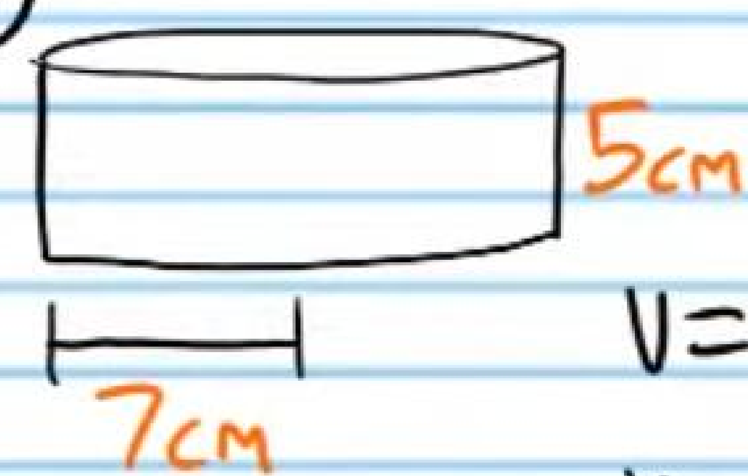
⑥ 

$V = \pi r^2 h$

$V = \pi(5)^2(18)$

$V = \pi(25)18$

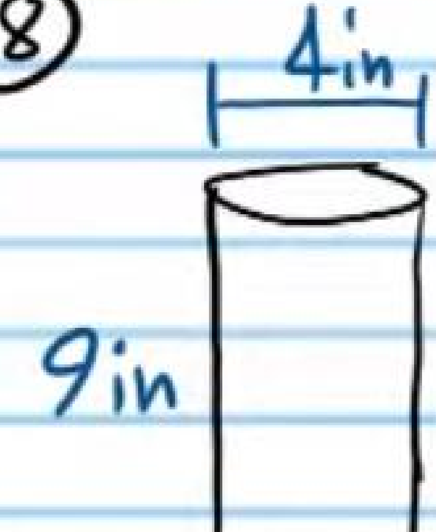
$\boxed{V = 450\pi\ in^3}$

⑦ 

$V = \pi(7)^2(5)$

$V = \pi(49)(5)$

$\boxed{V = 245\pi\ cm^3}$

⑧ 

$V = \pi(2)^2(9)$

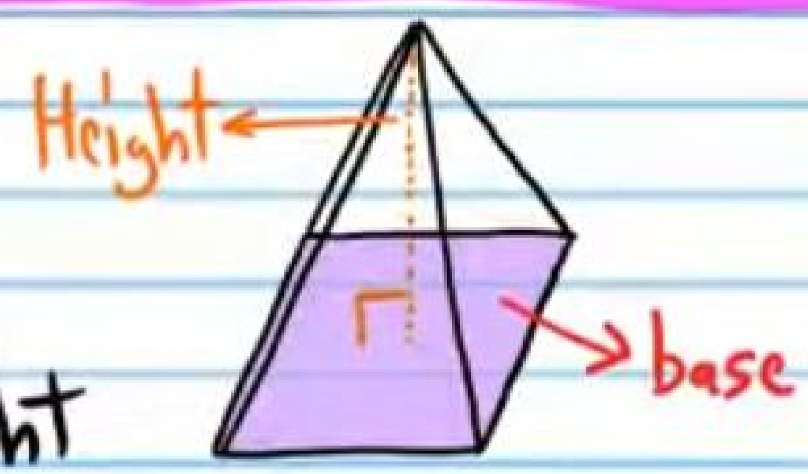
$V = \pi(4)(9)$

$\boxed{V = 36\pi\ in^3}$

Volume of a Pyramid

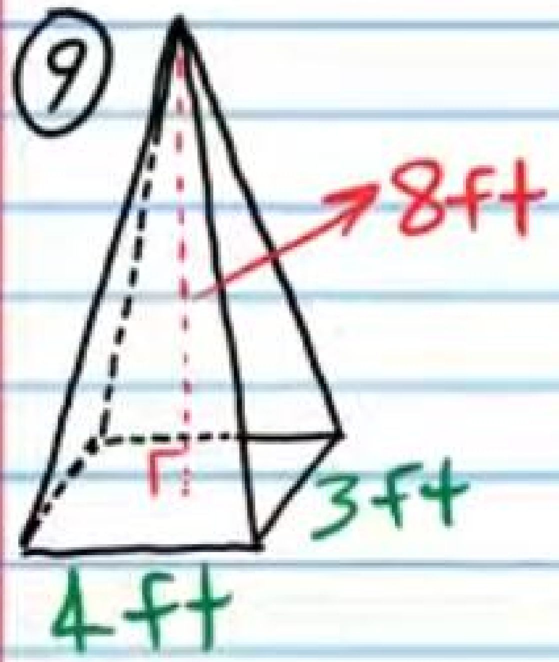
$V = \frac{1}{3}Bh$

B = base area & h = height



Find the volume of each pyramid below. Round to the nearest tenth.

⑨



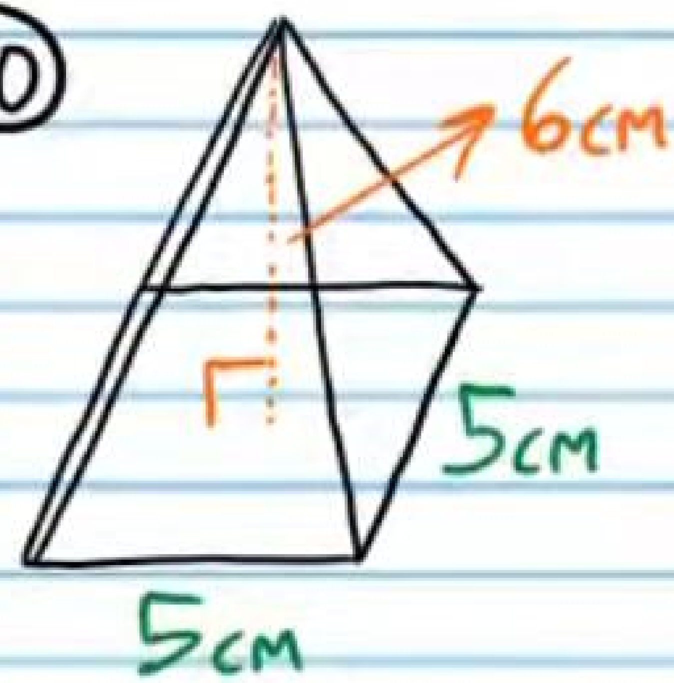
I) $B = 4(3) = 12$

II) $V = \frac{1}{3}Bh$

$V = \frac{1}{3}(12)(8)$

$\boxed{V = 32\,ft^3}$

⑩



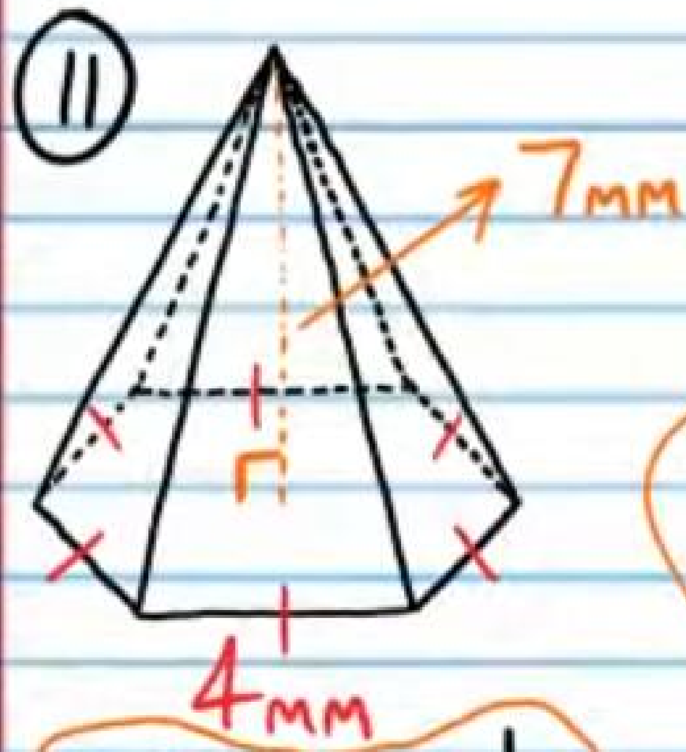
I) $B = 5(5) = 25$

II) $V = \frac{1}{3}Bh$

$V = \frac{1}{3}(25)(6)$

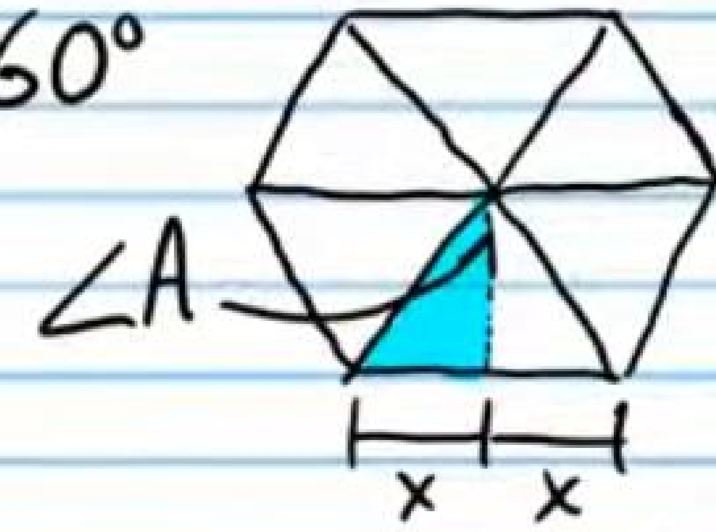
$\boxed{V = 50\,cm^3}$

⑪



I) Area of Polygon Base

Ia) $m\angle A = 360/6 = 60°$



Ib) Base

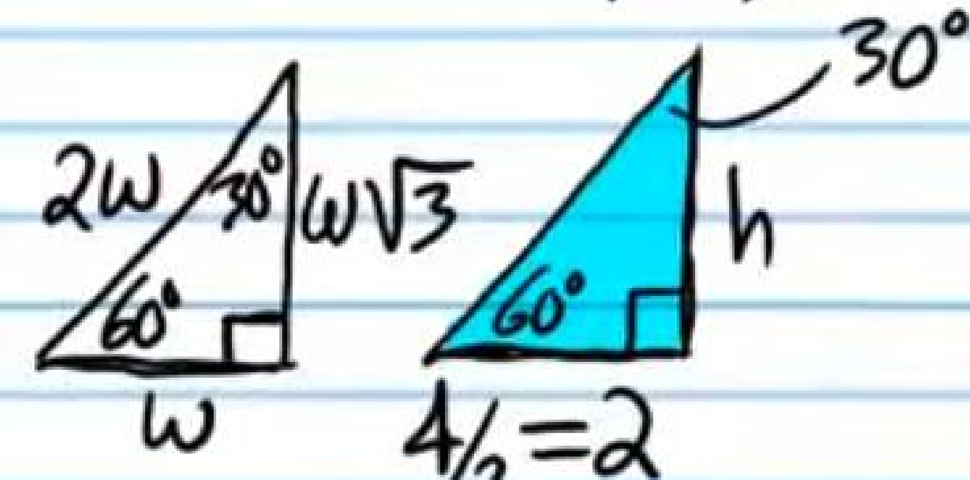
$b = 4$

Ic) Height

$w = 2$

$h = w\sqrt{3}$

$h = 2\sqrt{3}$



Id) Area of Triangle

$A = \frac{1}{2}bh$

$A = \frac{1}{2}(4)(2\sqrt{3})$

$A = 4\sqrt{3}$

Ie) Total area of Base

$A_{Tot} = 6A_{Tri}$

$A_{Tot} = 6(4\sqrt{3})$

$A_{Tot} = 24\sqrt{3}$

II) Volume

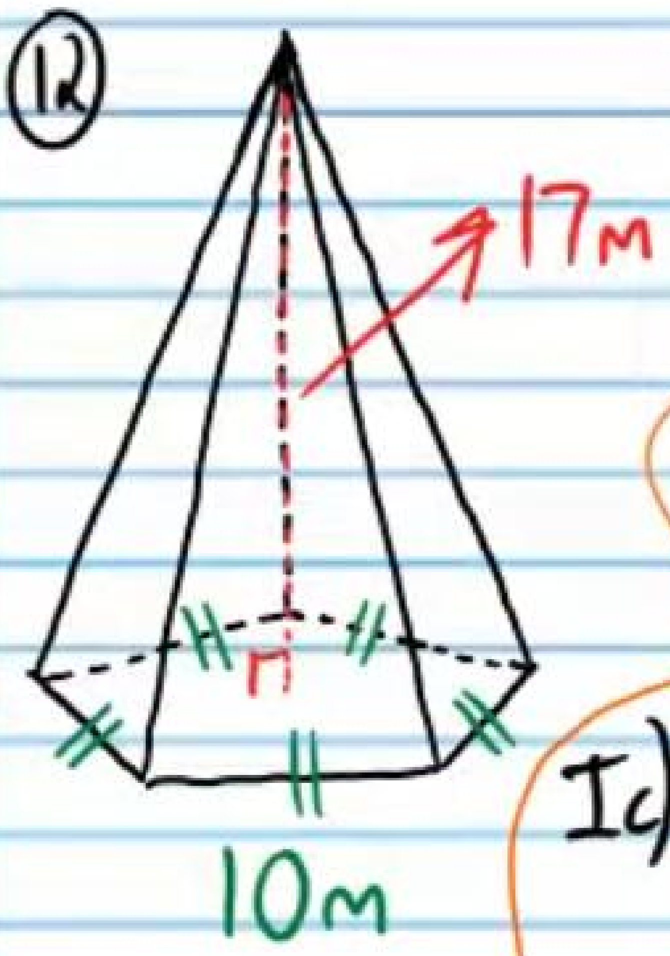
$B = 24\sqrt{3}$

$V = \frac{1}{3}Bh$

$V = \frac{1}{3}(24\sqrt{3})(7)$

$\boxed{V \approx 97\,mm^2}$

(12)



I) Area of Polygon Base

Ia) $m\angle A = 360/5 = 72^\circ$

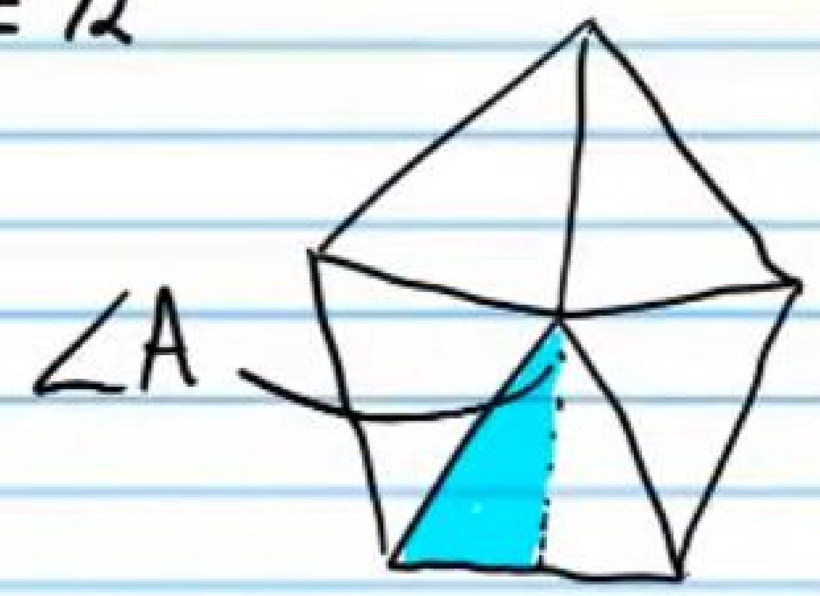
Ib) Base

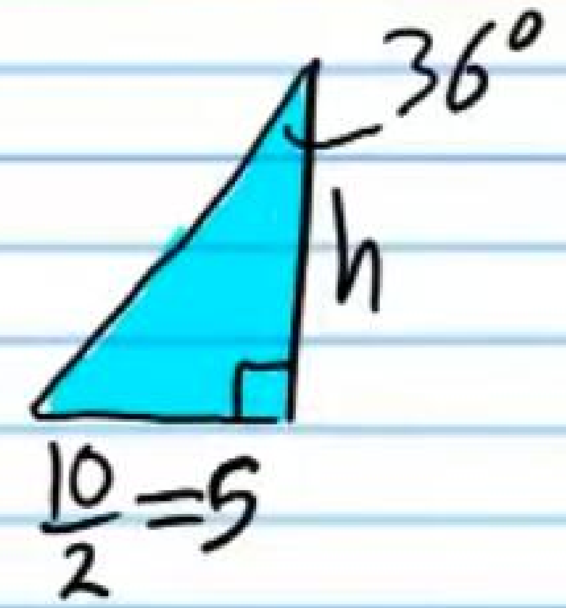
$b = 10$

Ic) Height

$\tan 36^\circ = \frac{5}{h}$

$h = \frac{5}{\tan 36^\circ}$





Id) Area of Triangle

$A = \frac{1}{2}bh$

$A = \frac{1}{2}(10)\left(\frac{5}{\tan 36^\circ}\right)$

Ie) Area of Base

$A_{Tot} = 5A_{Tri}$

$A_{Tot} = 5\left(\frac{1}{2}(10)\left(\frac{5}{\tan 36^\circ}\right)\right)$

II) Volume

$B = 5\left(\frac{1}{2}(10)\left(\frac{5}{\tan 36^\circ}\right)\right)$

$V = \frac{1}{3}Bh$

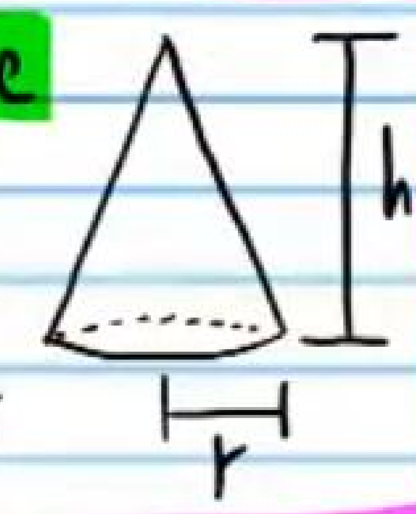
$V = \frac{1}{3}\left(5\left(\frac{1}{2}(10)\left(\frac{5}{\tan 36^\circ}\right)\right)\right)(17)$

$\boxed{V \approx 974.9\ m^3}$

Volume of a Cone

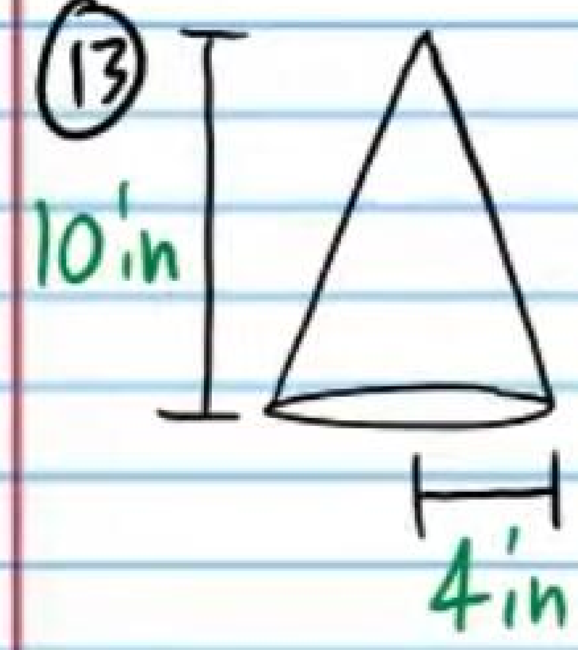
$V = \frac{1}{3}\pi r^2 h$

r = radius & h = height



Find the volume of each cone below. Round to the nearest thousandth.

(13)



$V = \frac{1}{3}\pi r^2 h$

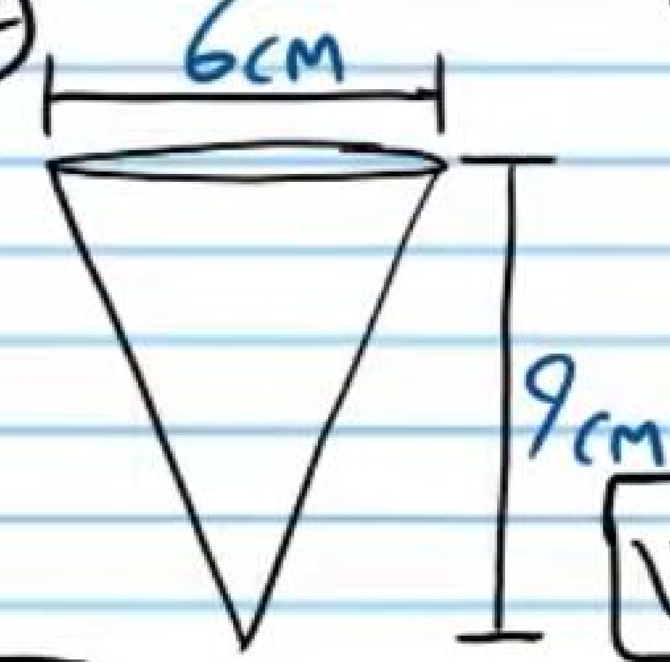
$V = \frac{1}{3}\pi(4)^2(10)$

$V = \frac{1}{3}\pi(16)(10)$

$V = \frac{1}{3}\pi(160)$

$\boxed{V \approx 167.552\ in^3}$

(14)



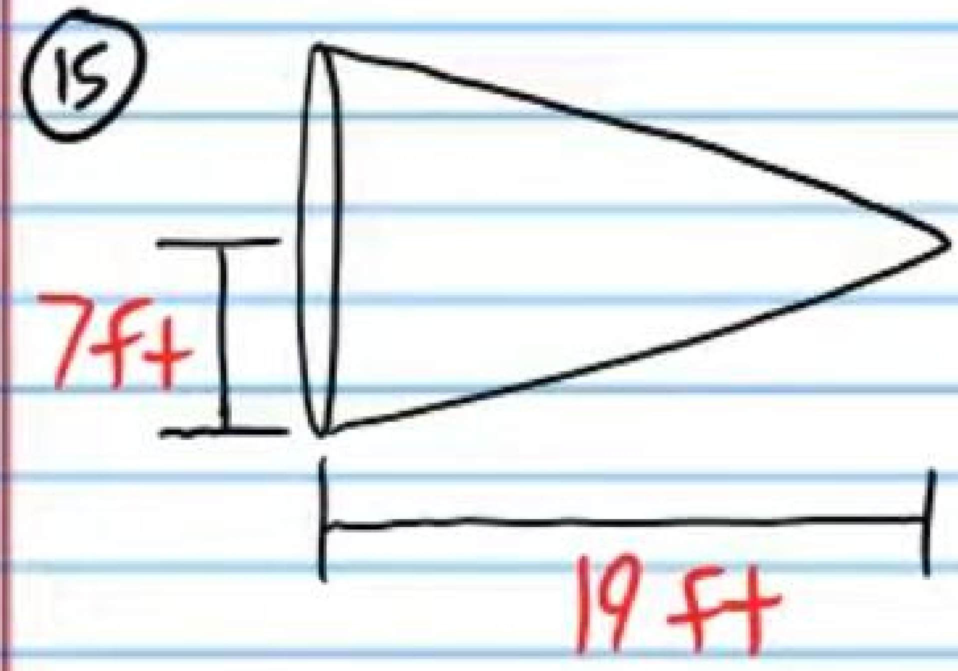
$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(3)^2(9)$

$V = 27\pi$

$\boxed{V \approx 84.823\ cm^3}$

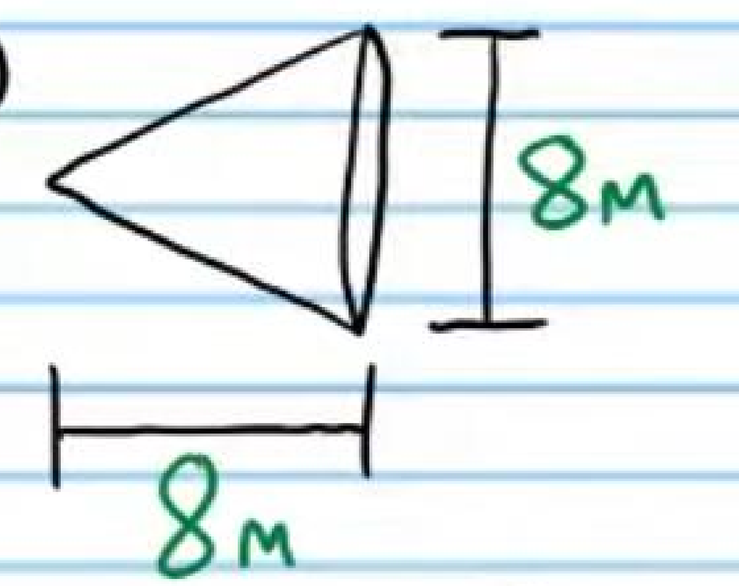
(15)



$V = \frac{1}{3}\pi(7)^2 19$

$\boxed{V \approx 974.941\ ft^3}$

(16)



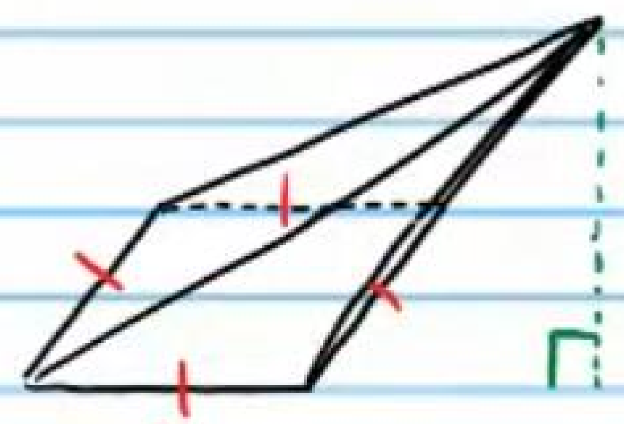
$V = \frac{1}{3}\pi(4)^2(8)$

$\boxed{V \approx 134.041\ m^3}$

More Problems

Find the volume of the following figures. Round to the nearest tenth if necessary.

(17)



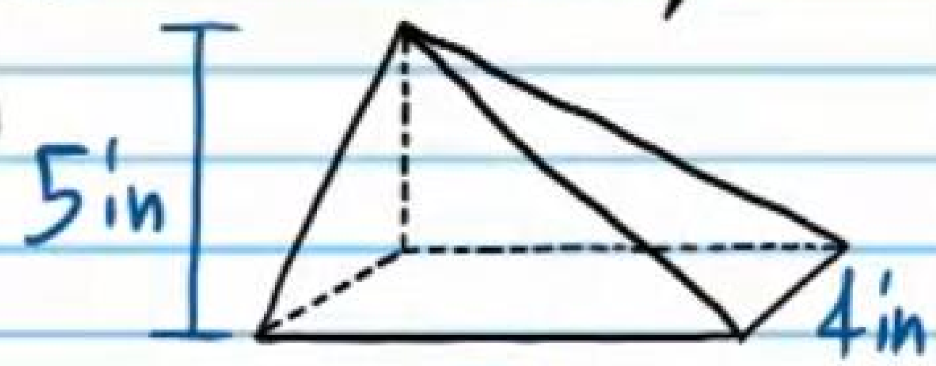
I) $B = 6(6) = 36$

II) $V = \frac{1}{3}Bh$

$V = \frac{1}{3}(36)9$

$\boxed{V = 108\ cm^3}$

(18)



I) $B = 7(4) = 28$

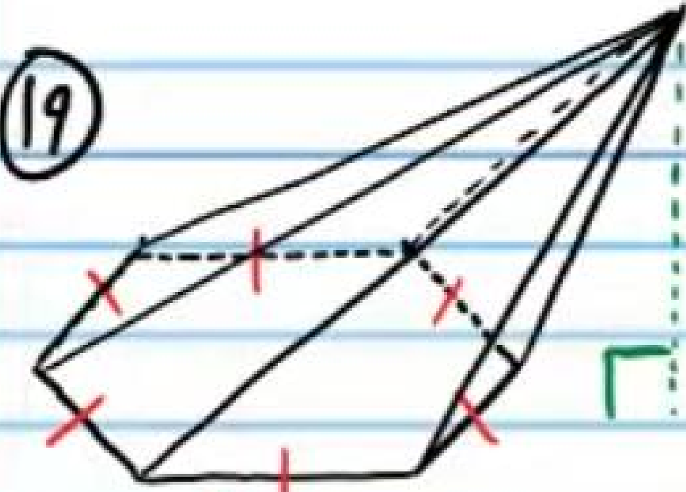
II) $V = \frac{1}{3}Bh$

$V = \frac{1}{3}(28)(5)$

$V = \frac{140}{3}$

$\boxed{V \approx 46.7\ in^3}$

(19)



I) Area of Polygon Base

* Refer to problem #11.

$B = 24\sqrt{3}$

II) Volume

$V = \frac{1}{3}Bh$

$V = \frac{1}{3}(24\sqrt{3})(6)$

$\boxed{V \approx 83.1\ mm^3}$

(20)

I) $B = \frac{1}{2}bh$

$B = \frac{1}{2}(5)(6)$

$B = 15$

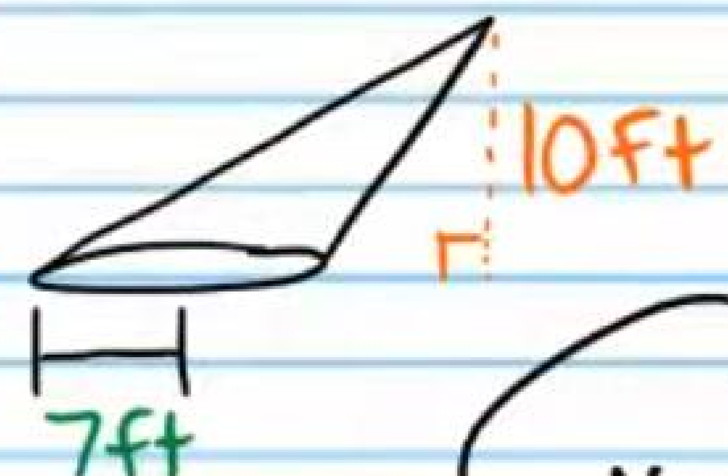
II) $V = \frac{1}{3}Bh$

$V = \frac{1}{3}(15)(4)$

$\boxed{V = 20\ m^3}$

Find the volume of the following figures. Leave your answers in terms of π.

(21)



$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(7)^2(10)$

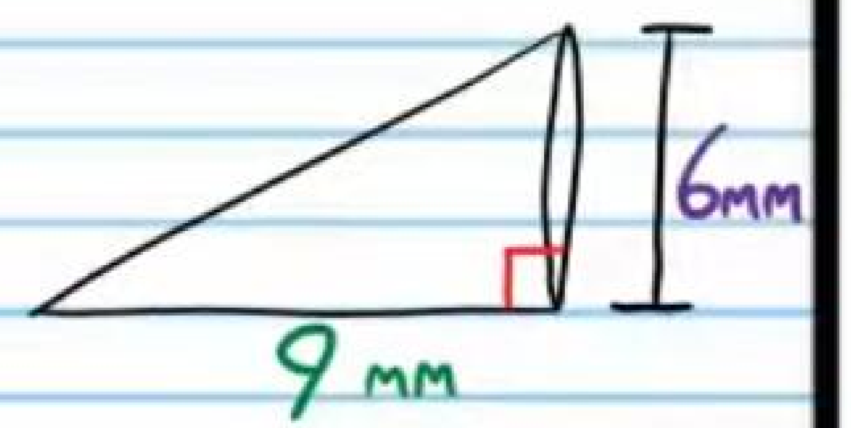
$V = \frac{1}{3}\pi(49)(10)$

$V = \frac{1}{3}\pi(490)$

$V = \frac{1}{3}(490\pi)$

$\boxed{V = \frac{490\pi}{3}\ ft^3}$

(22)

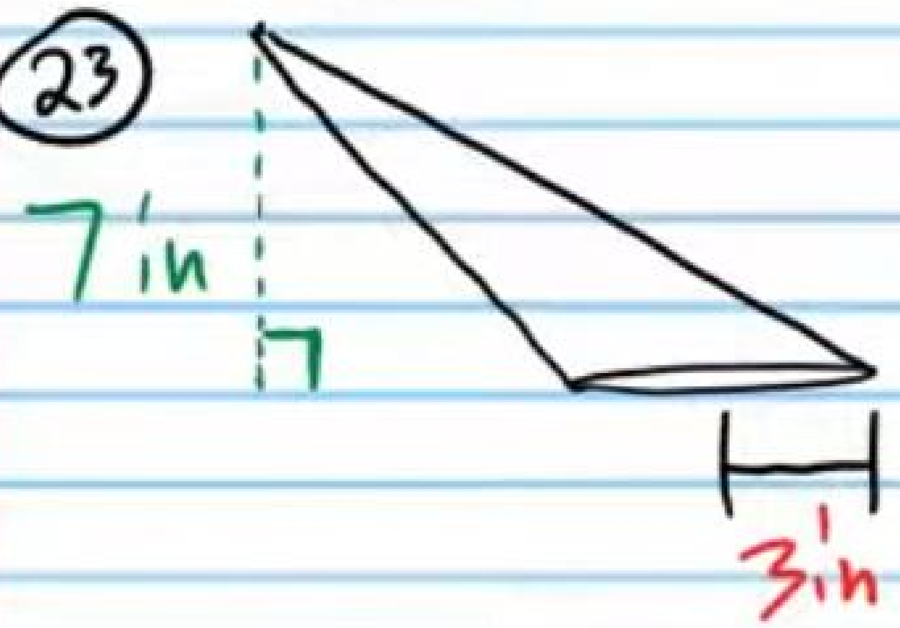


$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(3)^2(9)$

$\boxed{V = 27\pi\ mm^2}$

(23)

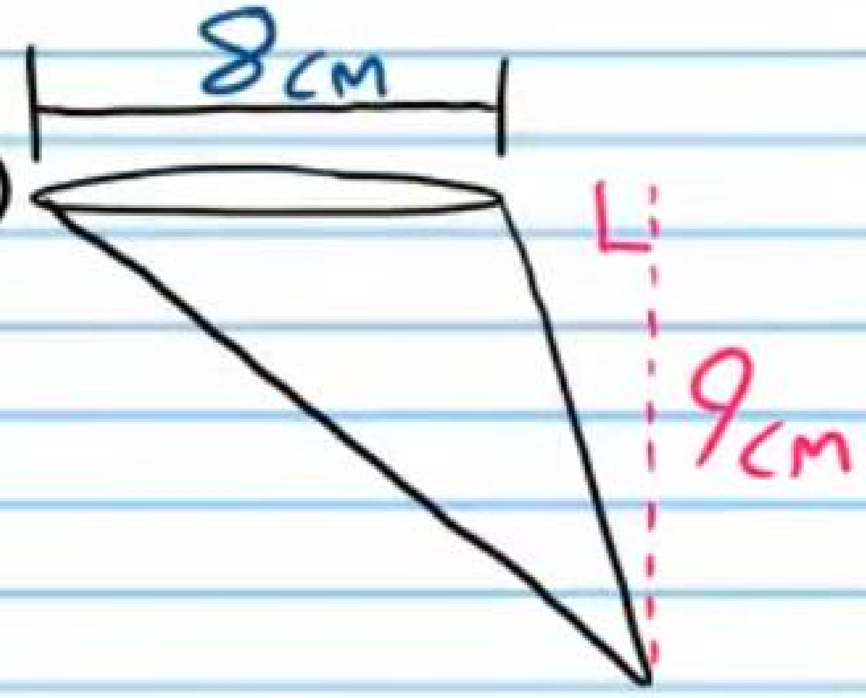


$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(3)^2(7)$

$\boxed{V = 21\pi\ in^3}$

(24)



$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(4)^2(9)$

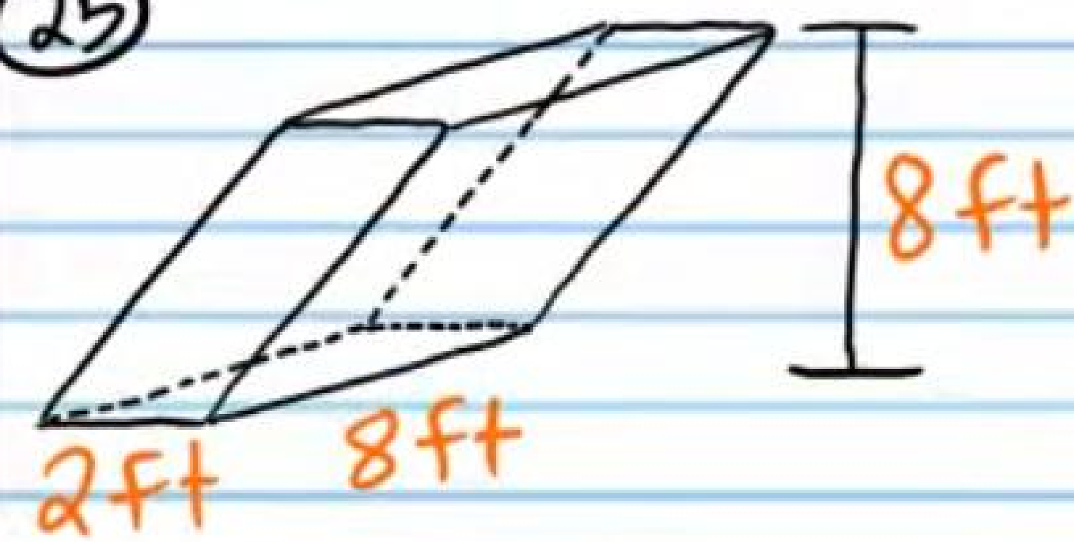
$\boxed{V = 48\pi\ cm^3}$

Cavalieri's Principle

If two solids have the same height and the same cross-sectional area at every level, then they have the same volume.

Find the volume of each figure below. For problem #'s 27–28, leave your answers as multiples of π.

(25)

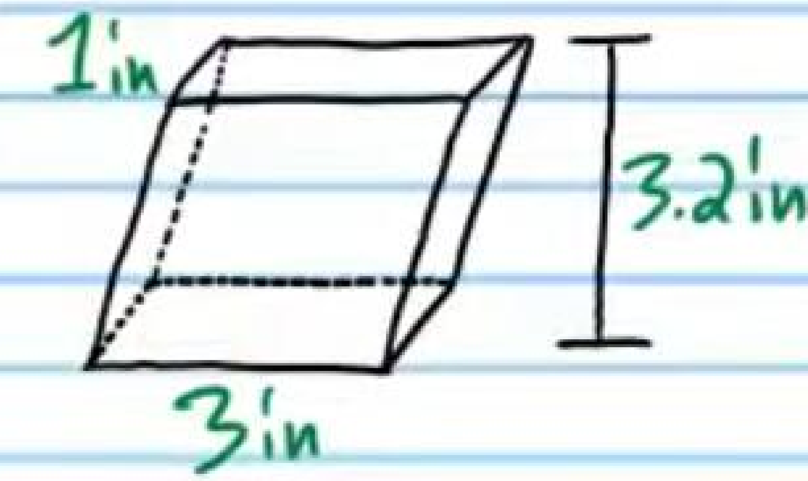


$V = LWH$

$V = 2(8)(8)$

$\boxed{V = 128\ ft^3}$

(26)

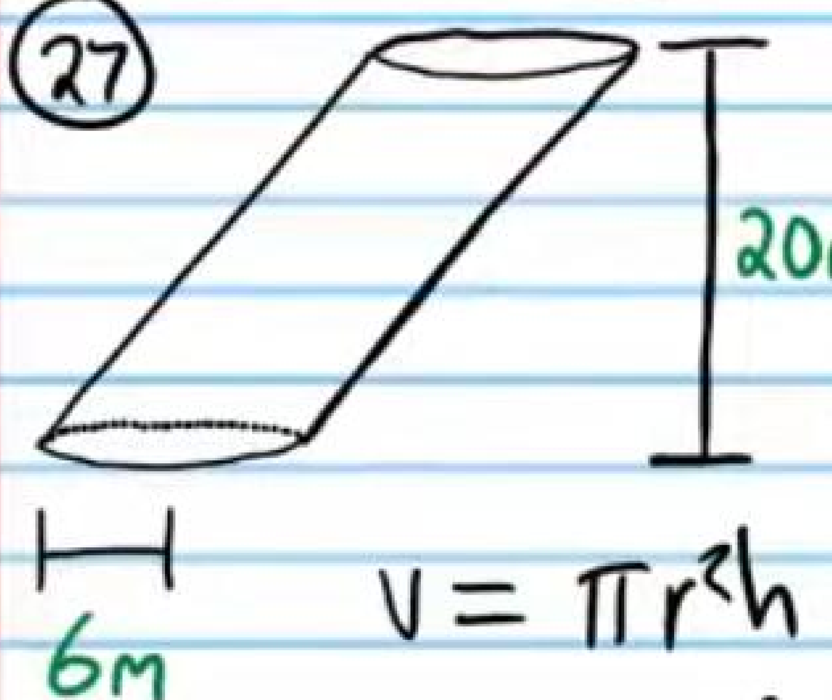


$V = LWH$

$V = 1(3)(3.2)$

$\boxed{V = 9.6\ in^3}$

(27)

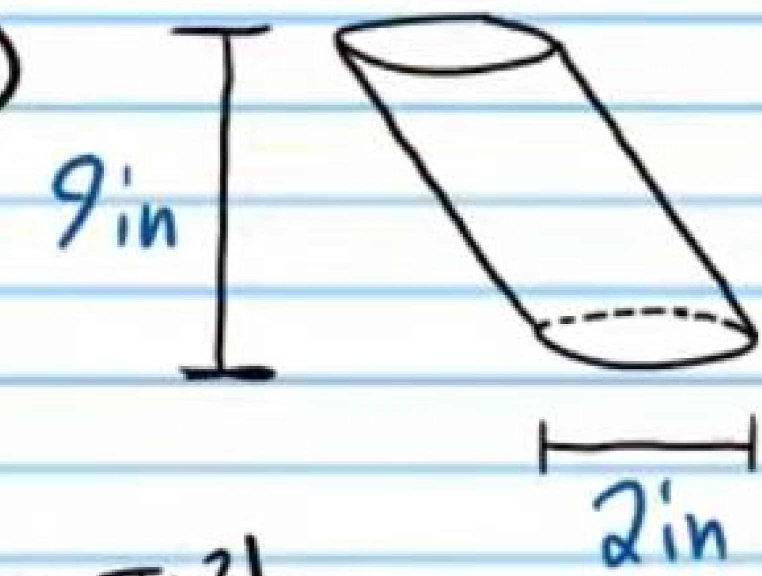


$V = \pi r^2 h$

$V = \pi(6)^2(20)$

$\boxed{V = 720\pi\ m^3}$

(28)



$V = \pi r^2 h$

$V = \pi(1)^2(9)$

$\boxed{V = 9\pi\ in^3}$

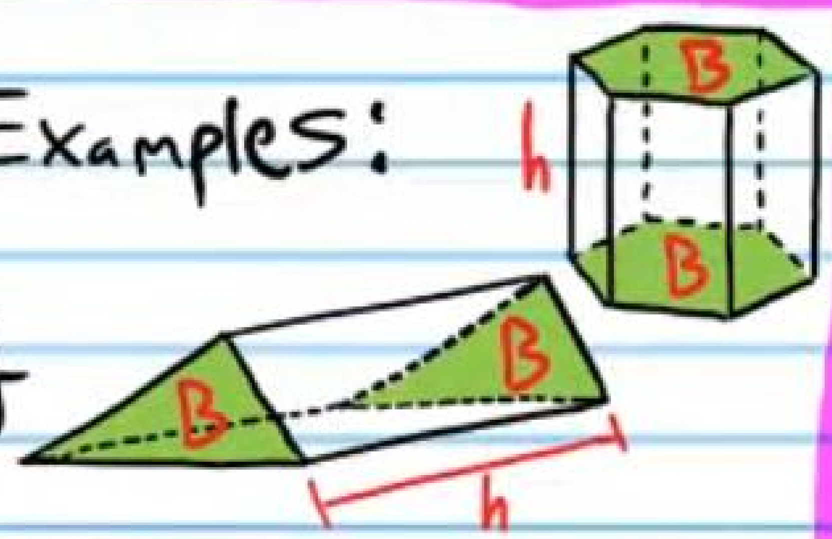
Prism: A three dimensional shape with two congruent and parallel sides called bases.

Volume of Any Prism

$$V = Bh$$

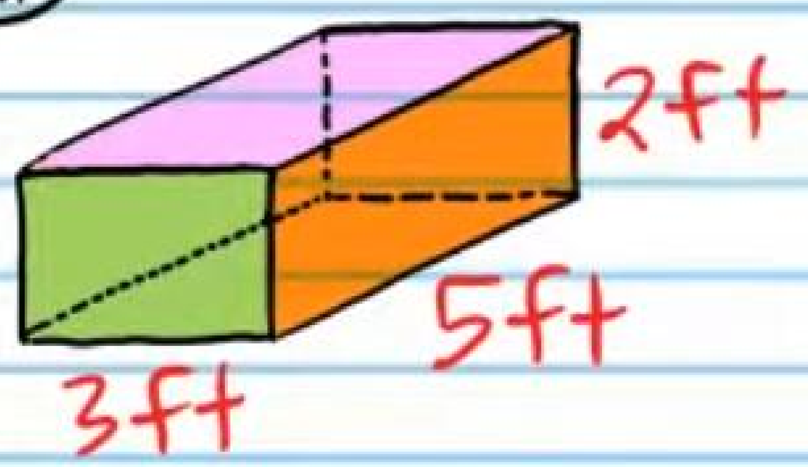
B = base area & h = height

Examples:



Find the volume of each prism below.

(29)

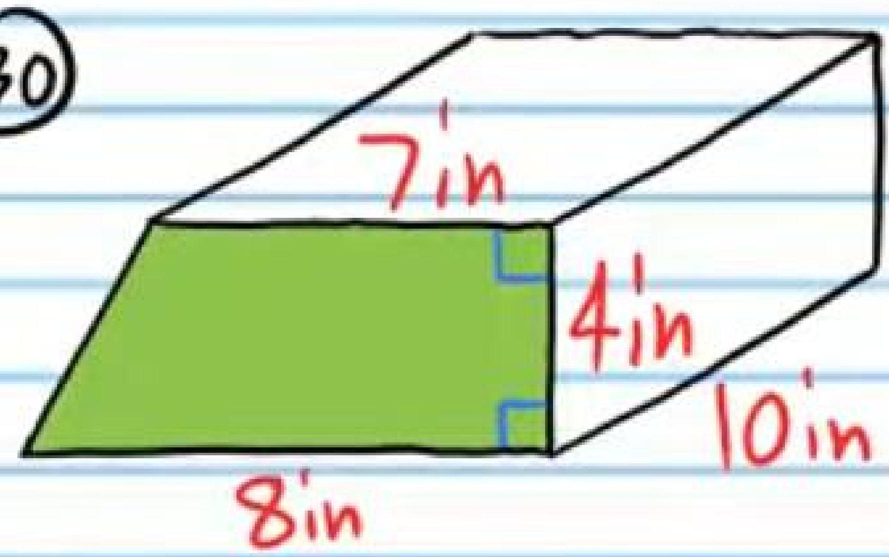


$V = Bh = LWH$

$= 3(5)(2)$

$= \boxed{30\ ft^3}$

(30)



I) $B = A_{Trapezoid} = \dfrac{h(b_1+b_2)}{2}$

$= \dfrac{4(7+8)}{2}$

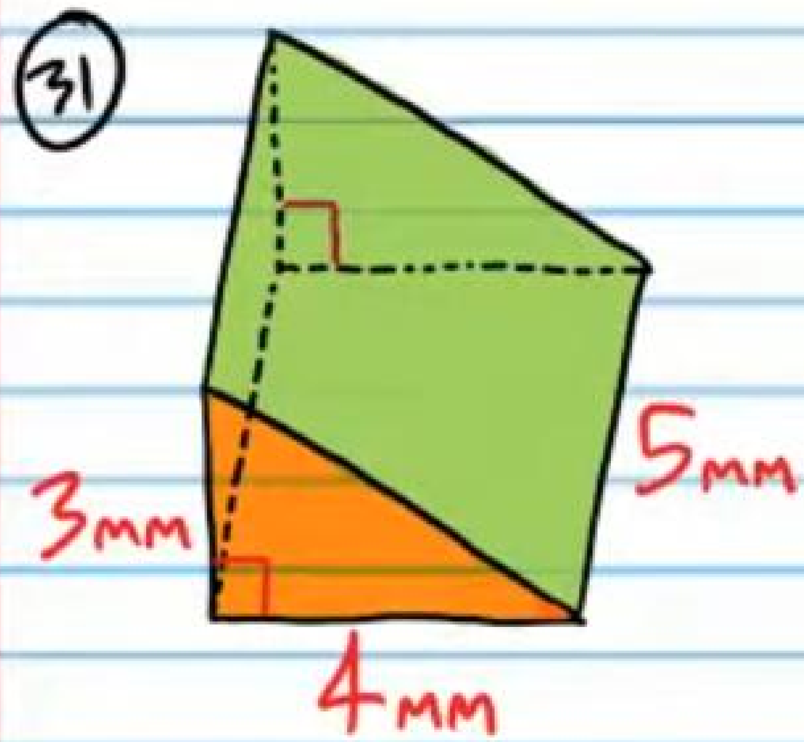
$= 30$

II) $V = Bh$

$V = (30)(10)$

$\boxed{V = 300\ in^3}$

(31)



I) $B = \frac{1}{2}bh$

$B = \frac{1}{2}(4)(3)$

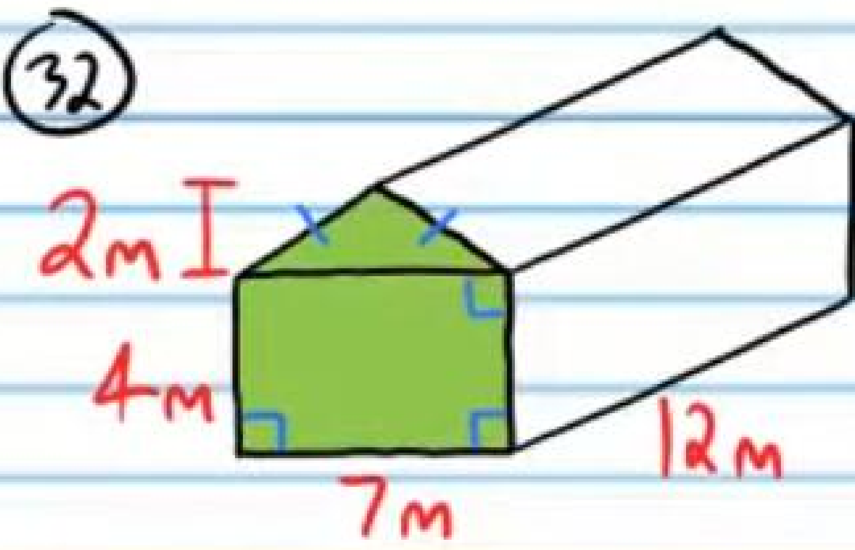
$B = 6$

II) $V = Bh$

$V = 6(5)$

$\boxed{V = 30\ mm^3}$

(32)



I) $B = A_{Rect} + A_{Tri}$

$B = LW + \frac{1}{2}bh$

$B = 4(7) + \frac{1}{2}(7)(2)$

$B = 28 + 7$

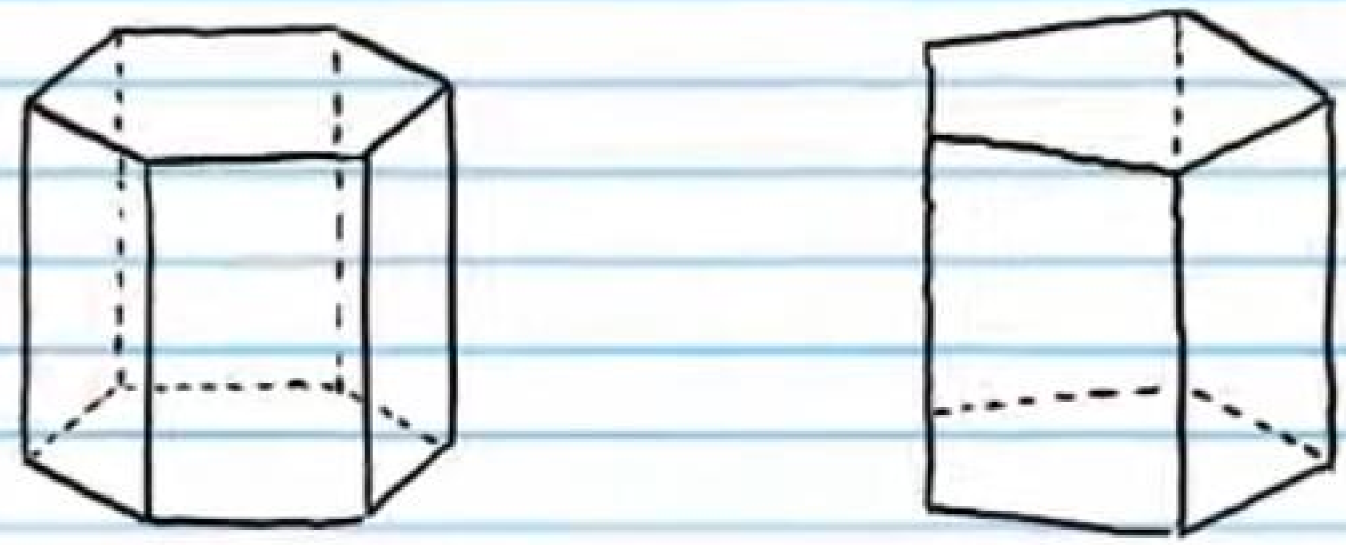
$B = 35$

II) $V = Bh$

$V = (35)(12)$

$\boxed{V = 420\ m^3}$

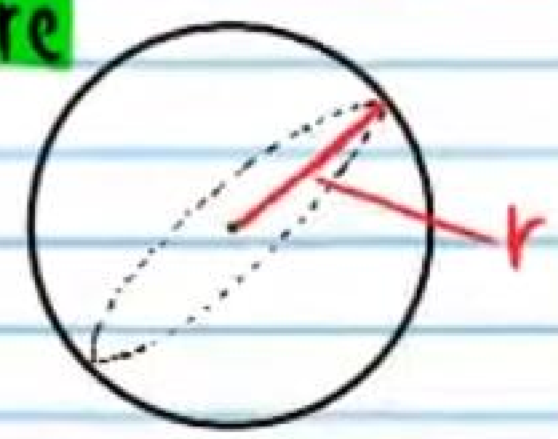
Other Examples of Prisms



Volume of a Sphere

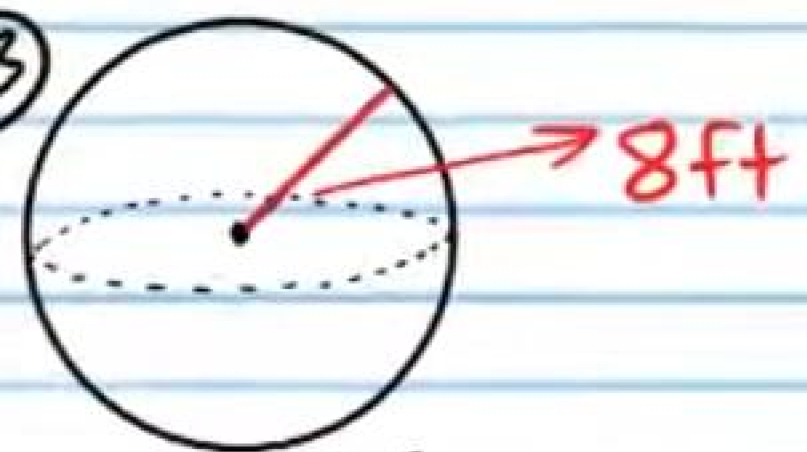
$V = \frac{4}{3}\pi r^3$

r = radius



Find the volume of each sphere below. Round to the nearest tenth.

33

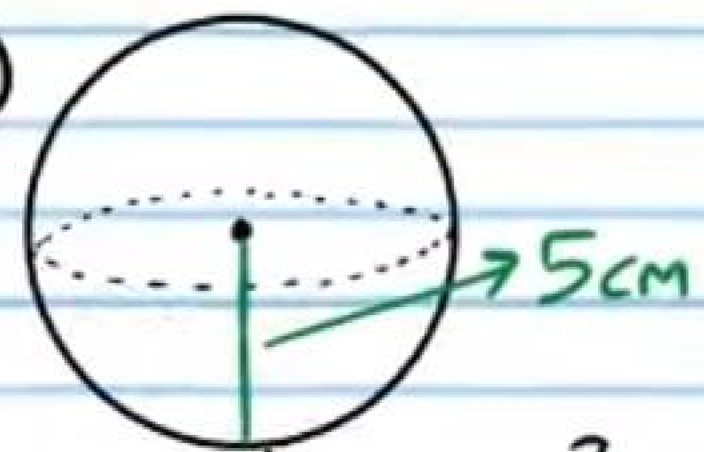


$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi (8)^3$

$\boxed{V \approx 2{,}144.7 \text{ ft}^3}$

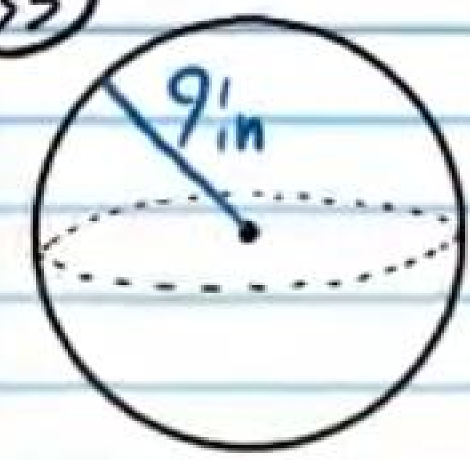
34



$V = \frac{4}{3}\pi (5)^3$

$\boxed{V \approx 523.6 \text{ cm}^3}$

35

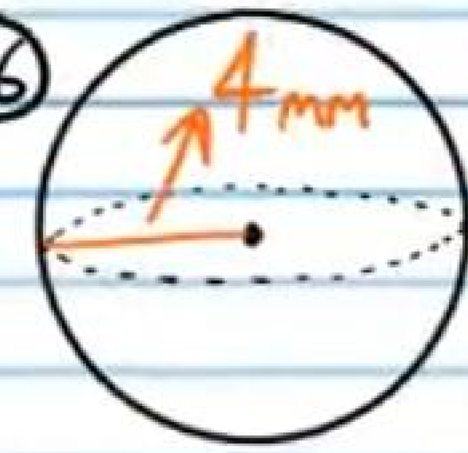


$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi (9)^3$

$\boxed{V \approx 3{,}053.6 \text{ in}^3}$

36



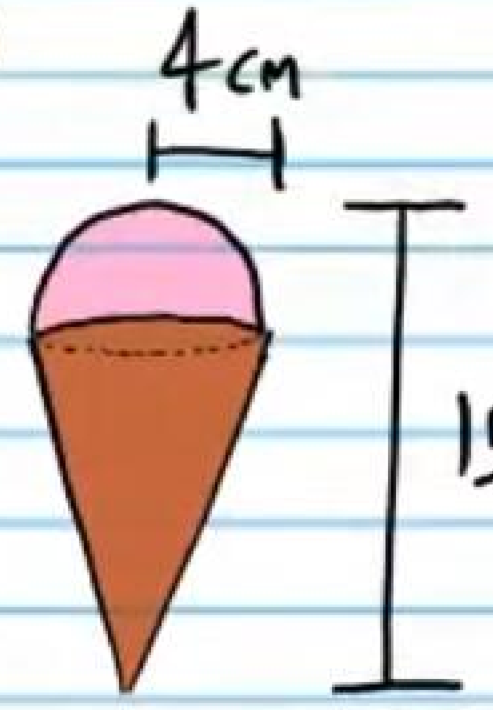
$V = \frac{4}{3}\pi (4)^3$

$\boxed{V \approx 268.1 \text{ mm}^3}$

Volume of Composite Figures

Find the volume of the following solids. Round to the nearest hundredth if necessary.

37



$$V = V_{cone} + V_{Hemi}$$

$$V = \frac{1}{3}\pi r^2 h + \frac{\frac{4}{3}\pi r^3}{2}$$

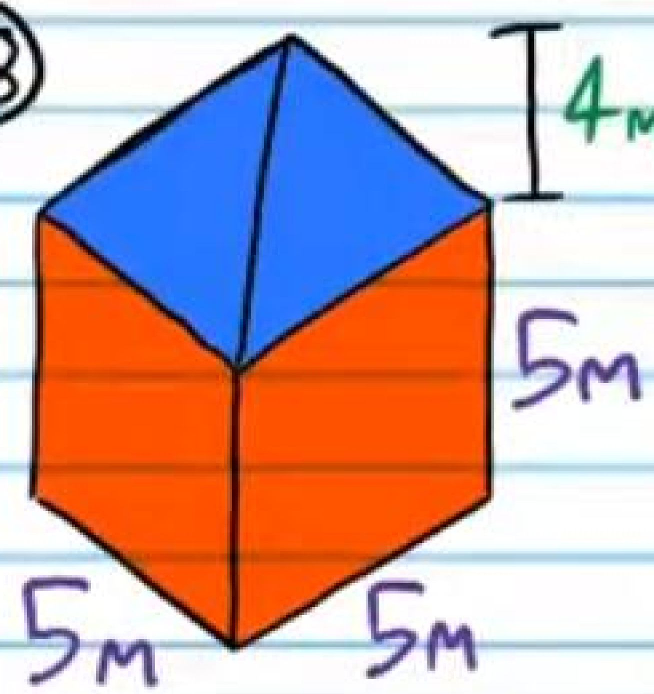
$$V = \frac{1}{3}\pi(4)^2(15-4) + \frac{\frac{4}{3}\pi(4)^3}{2}$$

$$V = \frac{1}{3}\pi(16)(11) + \frac{\frac{4}{3}\pi(64)}{2}$$

$$V \approx 184.306769 + 134.041286$$

$$\boxed{V \approx 318.35\text{ cm}^3}$$

38



$$V = V_{cube} + V_{pyr}$$

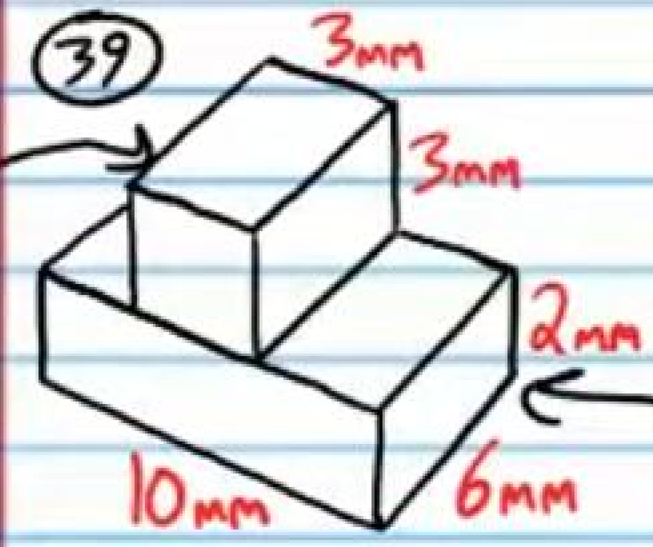
$$V = LWH + \frac{1}{3}Bh$$

$$V = 5(5)(5) + \frac{1}{3}(5)(5)(4)$$

$$V = 125 + \frac{100}{3}$$

$$\boxed{V = 158.33\text{ m}^3}$$

39



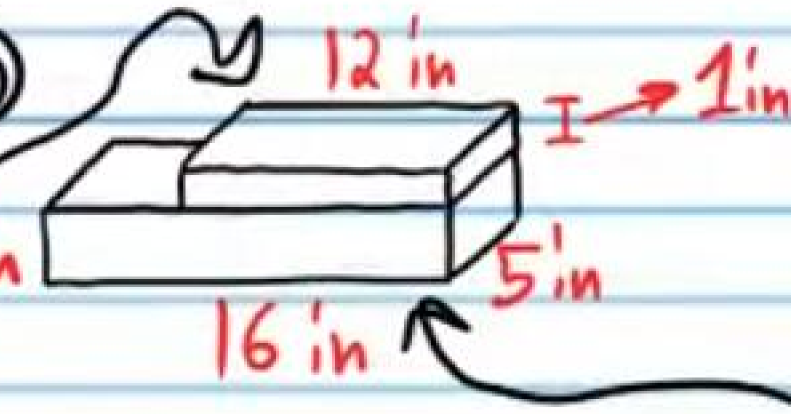
$$V = V_{Prism\,1} + V_{Prism\,2}$$

$$V = 3(3)(6) + 10(6)(2)$$

$$V = 54 + 120$$

$$\boxed{V = 174\text{ mm}^3}$$

40



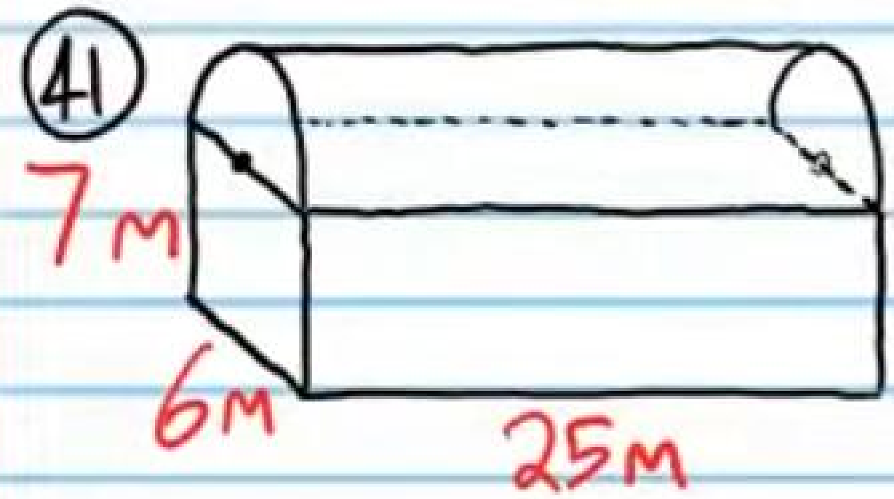
$$V = V_{Prism\,1} + V_{Prism\,2}$$

$$V = 1(12)(5) + 3(16)(5)$$

$$V = 60 + 240$$

$$\boxed{V = 300\text{ in}^3}$$

41



$$V = V_{Prism} + V_{Half\ cylinder}$$

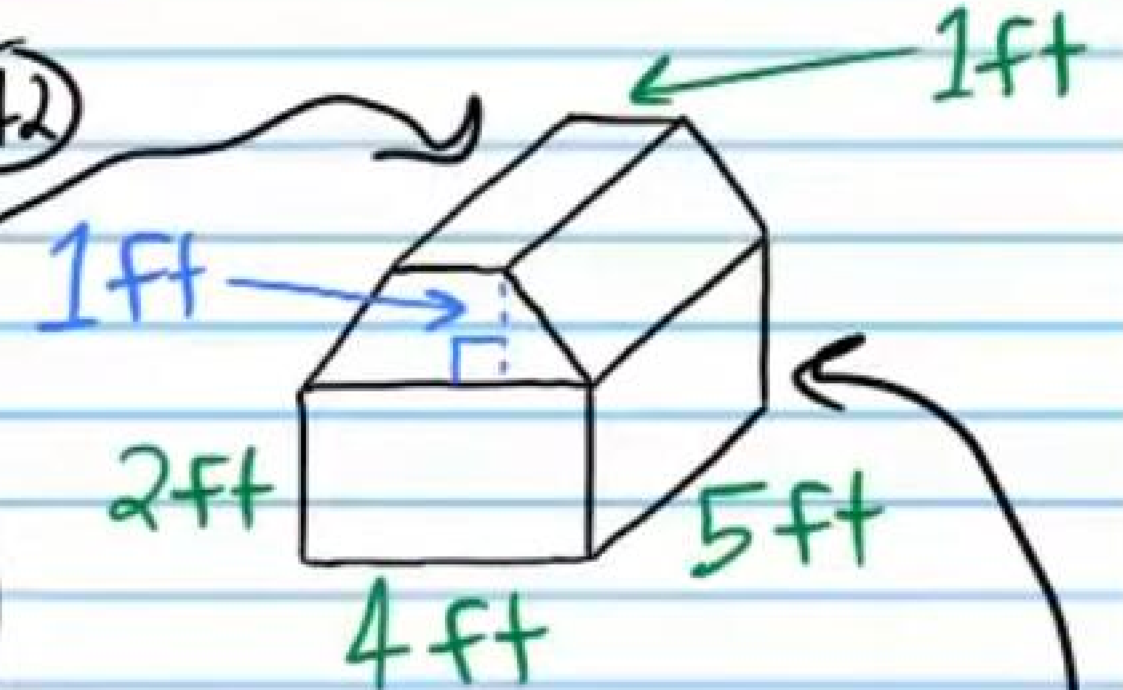
$$V = LWH + \frac{\pi r^2 h}{2}$$

$$V = 6(7)(25) + \frac{\pi(3)^2(25)}{2}$$

$$V \approx 1{,}050 + 353.429173$$

$$\boxed{V \approx 1{,}403.4\text{ m}^3}$$

42



$$V = V_{Prism\,1} + V_{Prism\,2}$$

$$V = Bh + LWH$$

$$V = \frac{1(1+4)}{2}(5) + 2(4)(5)$$

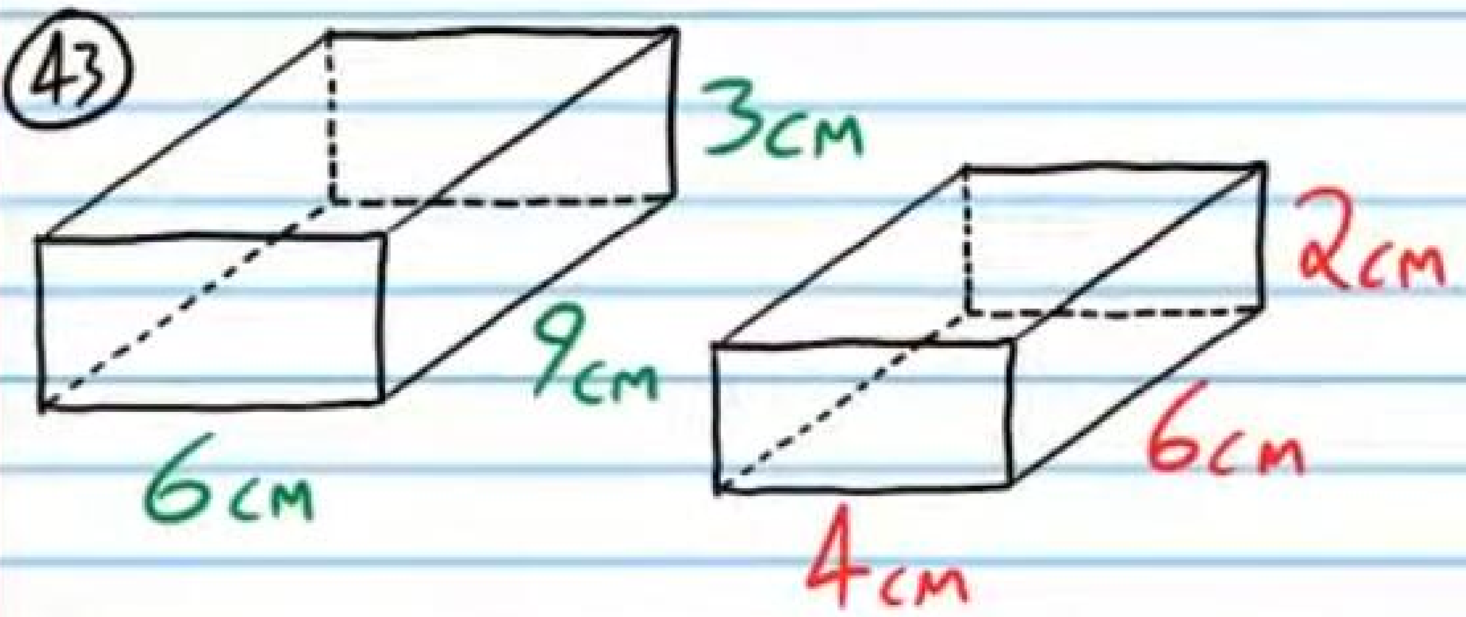
$$V = 12.5 + 40$$

$$\boxed{V = 52.5\text{ ft}^3}$$

Similar Solids: Solids with equal ratios of corresponding dimensions. As with two dimensional shapes, this ratio is called the scale factor.

Determine if each pair of figures below contains similar solids.

43

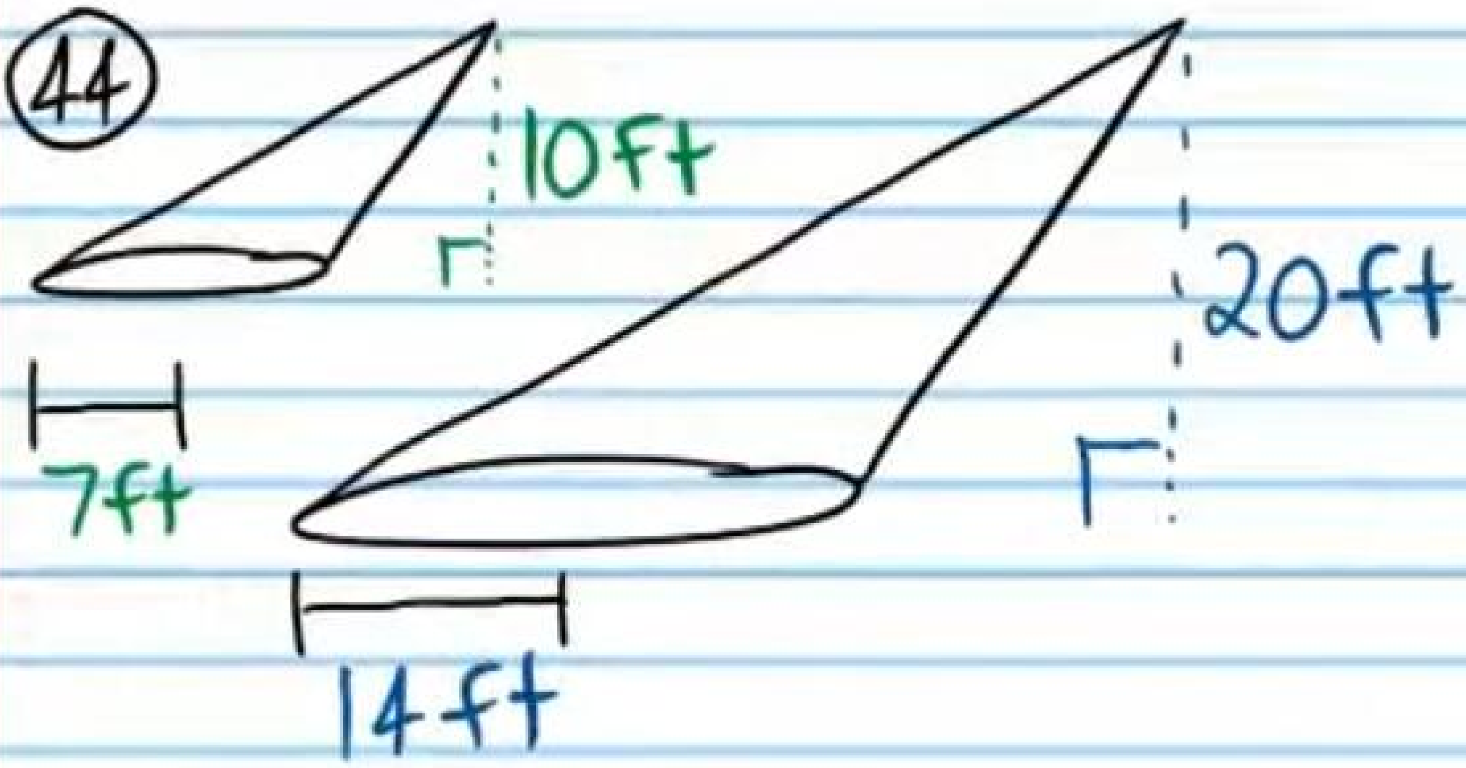


$$\frac{3}{2} = \frac{9}{6} = \frac{6}{4}$$

$$\frac{3}{2} = \frac{3}{2} = \frac{3}{2}$$

$\boxed{\text{Yes}}$

44

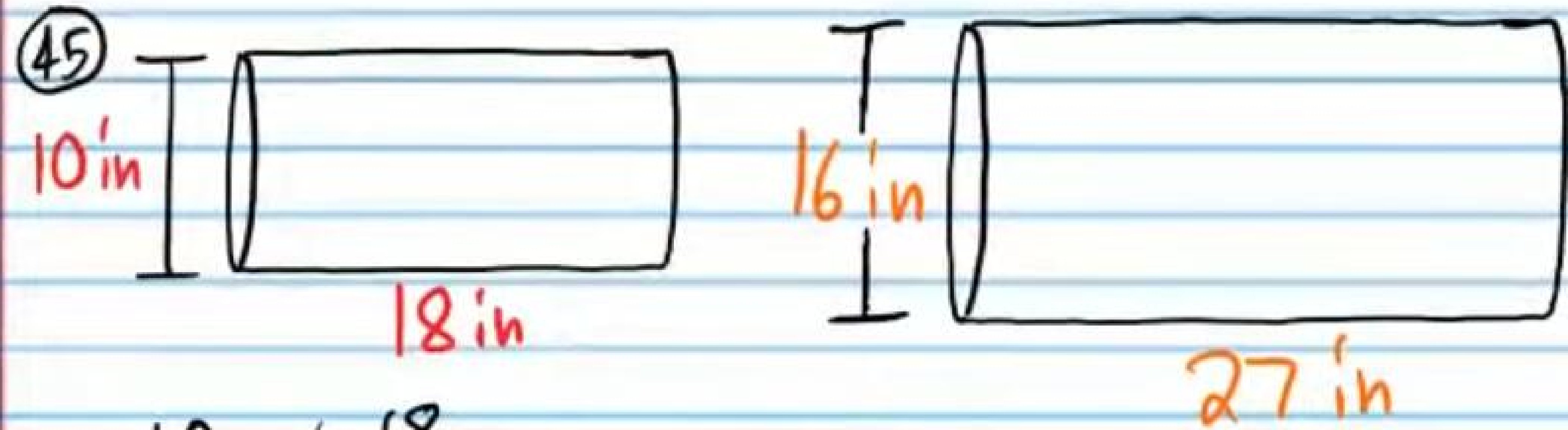


$$\frac{7}{14} = \frac{10}{20}$$

$$\frac{1}{2} = \frac{1}{2}$$

Yes

45

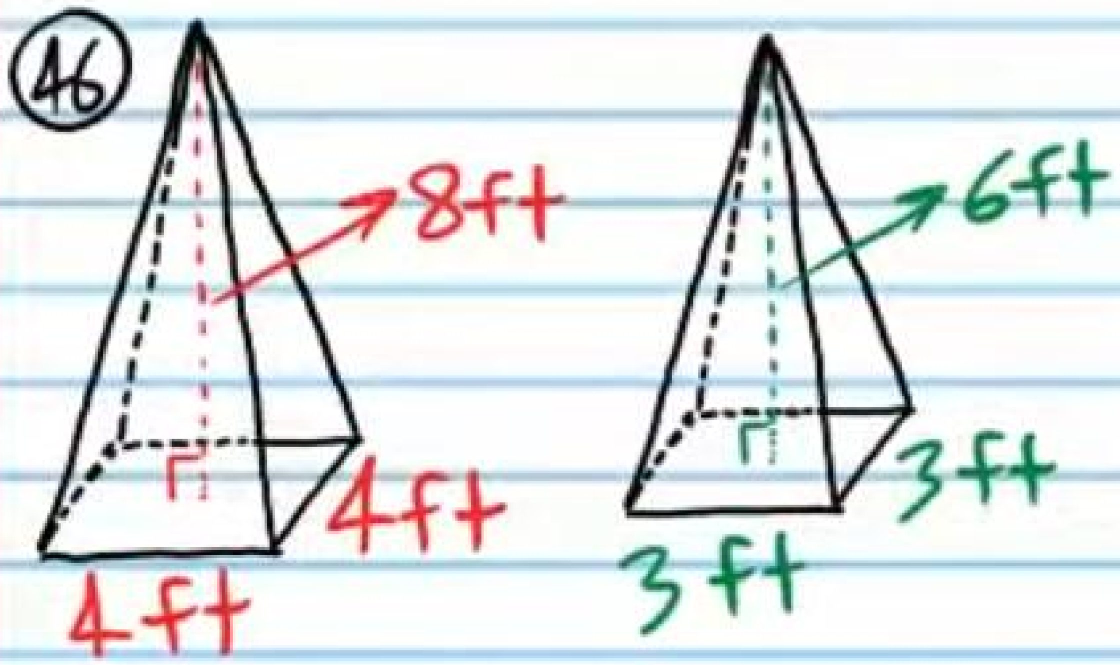


$$\frac{10}{16} \neq \frac{18}{27}$$

$$\frac{5}{8} \neq \frac{2}{3}$$

$\boxed{\text{NO}}$

46

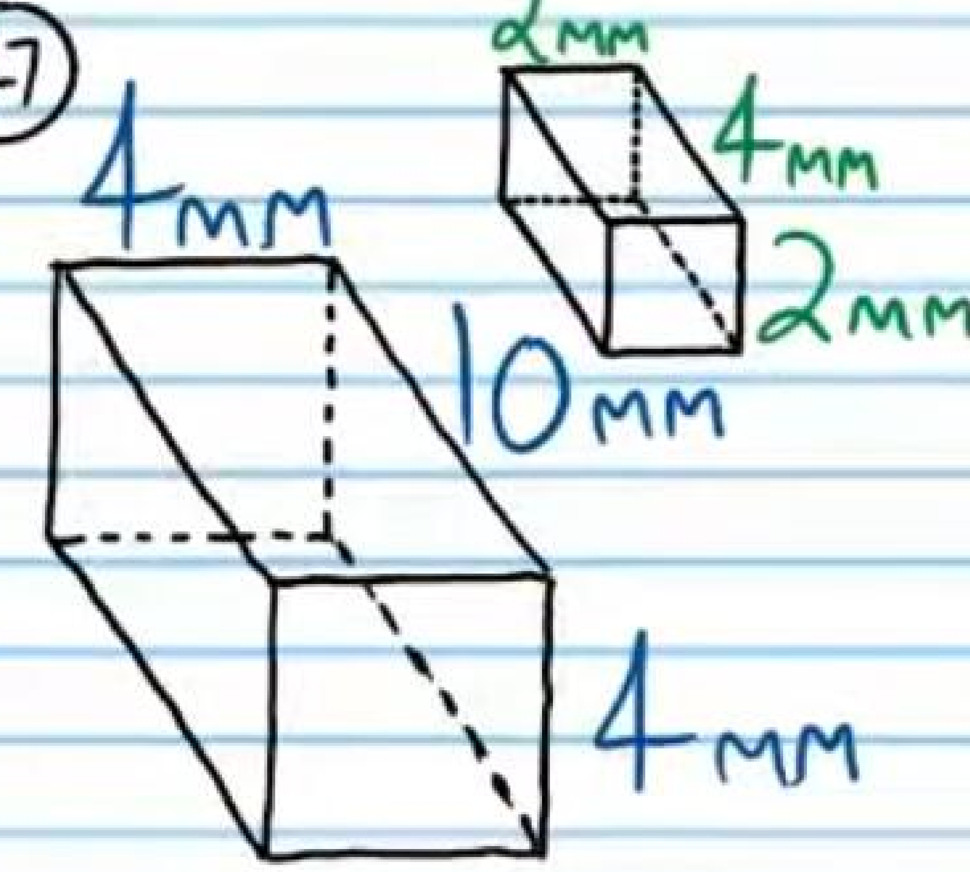


$$\frac{8}{6} = \frac{4}{3} = \frac{4}{3}$$

$$\frac{4}{3} = \frac{4}{3} = \frac{4}{3}$$

$\boxed{\text{Yes}}$

47



$$\frac{4}{2} = \frac{10}{4} = \frac{4}{2}$$

$$2 \neq 2.5 \neq 2$$

$\boxed{\text{NO}}$

Proportional Volumes Theorem: If two solids are similar, then the ratio of their volumes is equal to the cube of the ratio of their corresponding sides (i.e. the scale factor).

The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the volume of the smaller solid, find the volume of the larger solid.

48 $SF = 2/3,\ V = 40\ in^3$

$$\frac{40}{V_L} = \left(\frac{2}{3}\right)^3$$

$$\frac{40}{V_L} = \frac{2}{3}\cdot\frac{2}{3}\cdot\frac{2}{3}$$

$$\frac{40}{V_L} = \frac{8}{27}$$

$$1{,}080 = 8V_L$$

$$\boxed{V_L = 135\ in^3}$$

49 $SF = 4,\ V = 2\ m^3$

$$\frac{2}{V_L} = \left(\frac{1}{4}\right)^3$$

$$\frac{2}{V_L} = \frac{1}{4}\cdot\frac{1}{4}\cdot\frac{1}{4}$$

$$\frac{2}{V_L} = \frac{1}{64}$$

$$\boxed{V_L = 128\ m^3}$$

50 $SF = \frac{1}{2},\ V = 8\ cm^3$

$$\frac{8}{V_L} = \left(\frac{1}{2}\right)^3$$

$$\frac{8}{V_L} = \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}$$

$$\frac{8}{V_L} = \frac{1}{8}$$

$$\boxed{64\ cm^3 = V_L}$$

51 $SF = \frac{5}{4},\ V = 1\ ft^3$

$$\frac{1}{V_L} = \left(\frac{4}{5}\right)^3$$

$$\frac{1}{V_L} = \frac{64}{125}$$

$$125 = 64V_L$$

$$\boxed{\frac{125}{64}\ ft^3 = V_L}$$

The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the volume of the larger solid, find the volume of the smaller solid.

52 $SF = 3,\ V = 4\ mm^3$

$$\frac{4}{V_s} = \left(\frac{3}{1}\right)^3$$

$$\frac{4}{V_s} = \frac{27}{1}$$

$$4 = 27V_s$$

$$\boxed{\frac{4}{27}\ mm^3 = V_s}$$

53 $SF = \frac{1}{5},\ V = 10\ in^3$

$$\frac{V_s}{10} = \left(\frac{1}{5}\right)^3$$

$$\frac{V_s}{10} = \frac{1}{125}$$

$$125V_s = 10$$

$$\boxed{V_s = \frac{10}{125}\ in^3}$$

(54) $SF = \frac{1}{5},\ V = 8\,cm^3$

$$\frac{8}{V_s} = \left(\frac{5}{1}\right)^3$$

$$\frac{8}{V_s} = \frac{125}{1}$$

$$8 = 125V_s$$

$$\boxed{\frac{8}{125}\,cm^3 = V_s}$$

(55) $SF = 2,\ V = 2\,m^3$

$$\frac{V_s}{2} = \left(\frac{1}{2}\right)^3$$

$$\frac{V_s}{2} = \frac{1}{8}$$

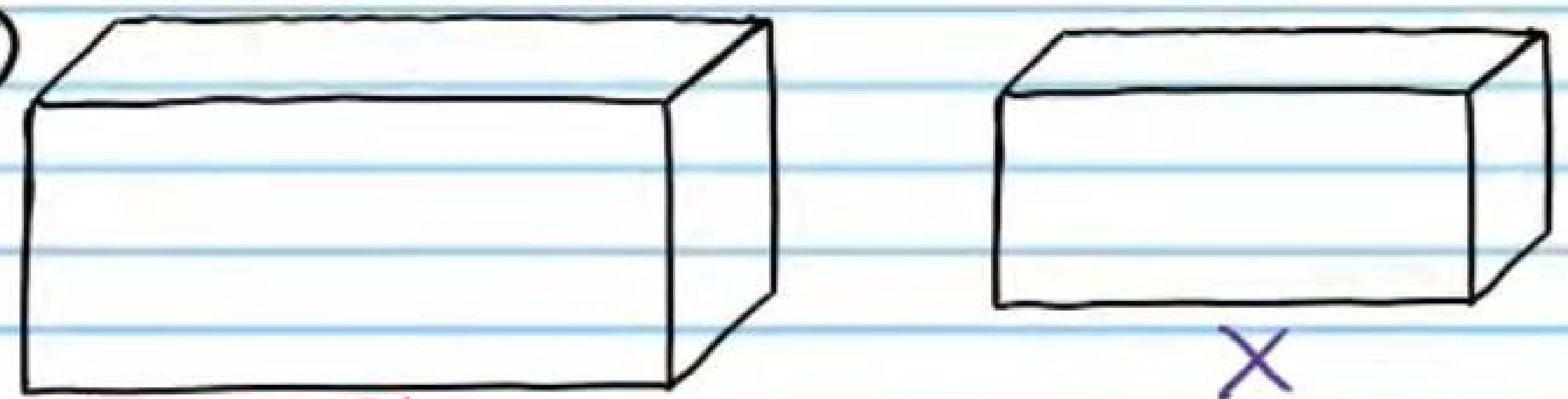
$$8V_s = 2$$

$$V_s = 2/8$$

$$\boxed{V_s = \tfrac{1}{4}\,m^3}$$

If the following pairs of figures below are similar solids, find the unknown values.

(56)



$V_2 = 33.75\,ft^3$

$V_1 = 80\,ft^3$

$$\frac{80}{33.75} = \left(\frac{8}{X}\right)^3$$

$$\sqrt[3]{\frac{80}{33.75}} = \sqrt[3]{\left(\frac{8}{X}\right)^3}$$

$$\sqrt[3]{\frac{80}{33.75}} = \frac{8}{X}$$

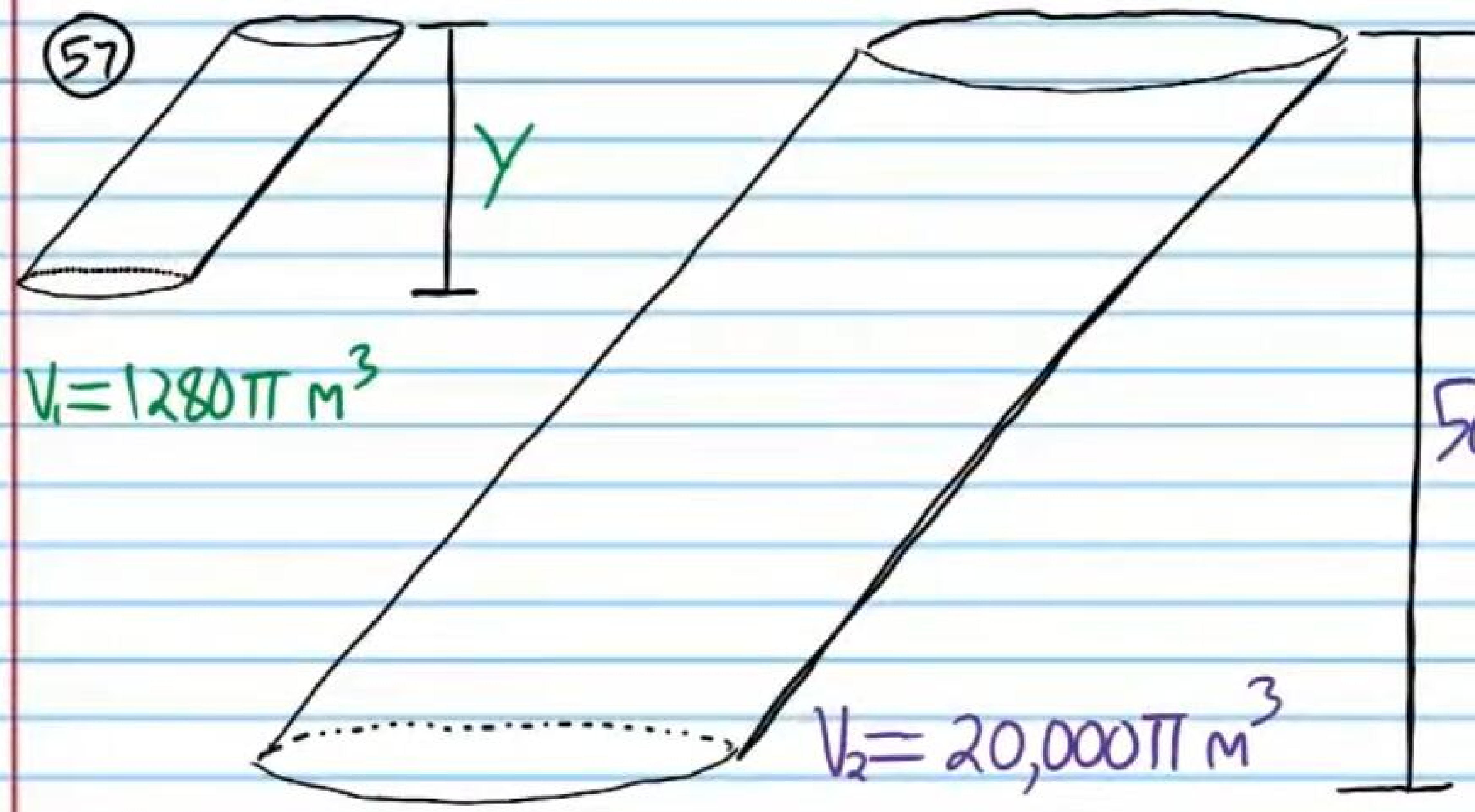
$$\frac{1.\overline{3}}{1} = \frac{8}{X}$$

$$1.\overline{3}X = 8$$

$$X = \frac{8}{1.\overline{3}}$$

$$\boxed{X = 6\,ft}$$

(57)



$$\frac{1280\pi}{20{,}000\pi} = \left(\frac{Y}{50}\right)^3$$

$$\sqrt[3]{\frac{1280}{20{,}000}} = \sqrt[3]{\left(\frac{Y}{50}\right)^3}$$

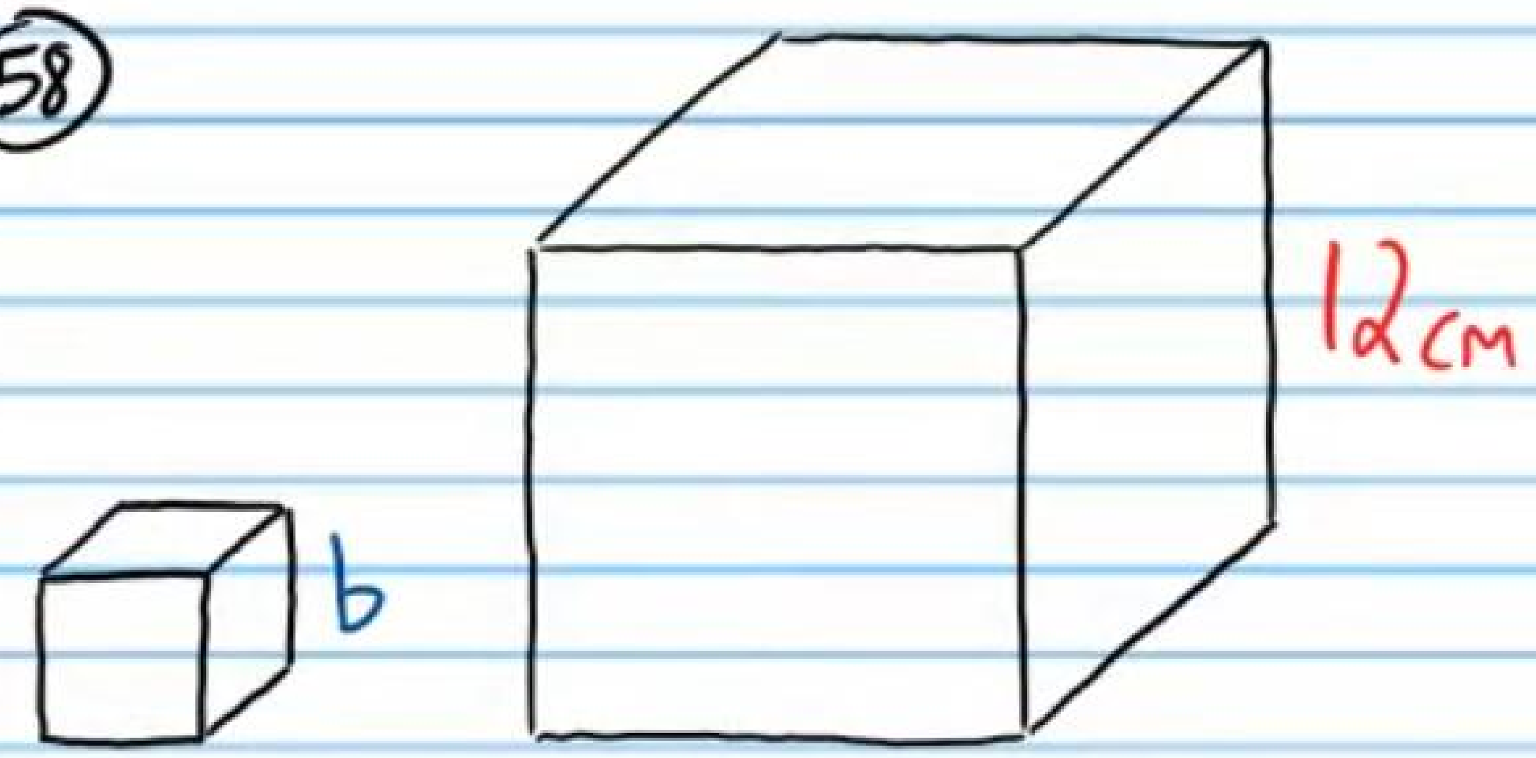
$$\boxed{\sqrt[Y]{X}}$$

$$\frac{0.4}{1} = \frac{Y}{50}$$

$$0.4(50) = Y$$

$$\boxed{20\,m = Y}$$

(58)



$V_1 = 64\,cm^3$ $\quad V_2 = 1{,}728\,cm^3$

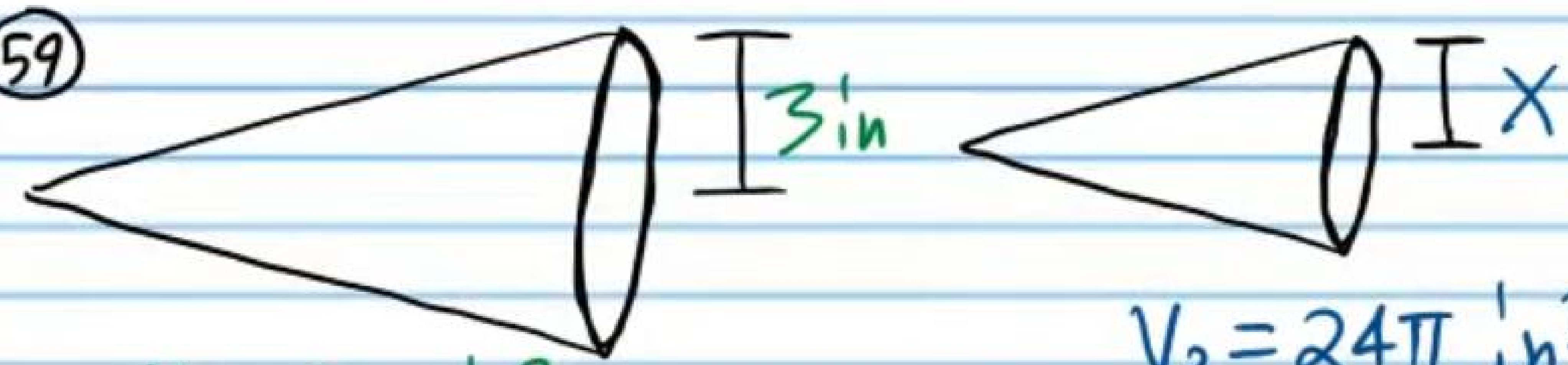
$$\frac{64}{1728} = \left(\frac{b}{12}\right)^3$$

$$\sqrt[3]{\frac{64}{1728}} = \sqrt[3]{\left(\frac{b}{12}\right)^3}$$

$$\frac{4}{12} = \frac{b}{12}$$

$$\boxed{b = 4\,cm}$$

(59)



$V_1 = 81\pi\,in^3$ $\quad V_2 = 24\pi\,in^3$

$$\frac{81\pi}{24\pi} = \left(\frac{3}{x}\right)^3$$

$$\sqrt[3]{\frac{27}{8}} = \sqrt[3]{\left(\frac{3}{x}\right)^3}$$

$$\frac{3}{2} = \frac{3}{x}$$

$$\boxed{x = 2\,in}$$